Vol 11 Chapter 7 — Heegner Numbers and Cremona Elliptic Curves

Keeper (author pass — deep math/physics revision)

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Chapter 7 — Heegner Numbers and Cremona Elliptic Curves

Level 1 — one sentence

Heegner-Stark-Baker (1952-1967, L1 ESTABLISHED) proved exactly nine imaginary quadratic fields have class number 1 (Heegner numbers $\{-1, -2, -3, -7, -11, -19, -43, -67, -163\}$), with BST anchoring on the small-primary subset $\{-3, -7, -11\}$ (K75 audit), and the elliptic curve Cremona 49a1 ($Y^2 = X^3 - 945X - 10206$) being BST's canonical curve with every invariant a BST primary (conductor $g^2 = 49$, j-invariant $-(N_c n_C)^3$, CM by $\mathbb{Q}(\sqrt{-g})$).

Level 2 — graduate-physicist precision

7.1 The nine Heegner numbers

Heegner 1952 / Stark 1967 / Baker 1966 (independent proofs): the only imaginary quadratic fields $\mathbb{Q}(\sqrt{-d})$ with class number 1 are

$$d \in \{1, 2, 3, 7, 11, 19, 43, 67, 163\}$$

(commonly listed with negative sign: $-1, -2, \ldots, -163$).

Significance: in these fields, every ideal is principal — unique factorization of ideals reduces to unique factorization of elements (in the integer ring of the field).

7.2 BST anchors on small-primary subset {-3, -7, -11}

K75 audit (Spring 2026): BST anchors structurally on the small-primary subset $\{-3, -7, -11\}$ of Heegner numbers — these match BST primaries: - $-3 = -N_c$: anchored via Cremona 27a1 (Task #186) - $-7 = -g$: anchored via Cremona 49a1 (the canonical curve, this chapter) - -11: special role, fewer immediate BST anchors but in pattern

The non-small-primary Heegners $\{-19, -43, -67, -163\}$ play different roles (or no immediate role) in BST.

7.3 Cremona 49a1: BST canonical curve

The elliptic curve labeled "49a1" in Cremona's database:

$$E : Y^2 = X^3 - 945X - 10206$$

Every invariant is BST-primary: - Conductor: $N = 49 = g^2 = 7^2$ - Discriminant: $\Delta = -343 = -g^3 = -7^3$ - j-invariant: $j(E) = -3375 = -15^3 = -(N_c \cdot n_C)^3$ - Torsion: $\mathbb{Z}/2\mathbb{Z}$ (rank 0 plus 2-torsion); torsion = rank = 2 in extended sense - Complex multiplication: by $\mathbb{Q}(\sqrt{-7}) = \mathbb{Q}(\sqrt{-g})$

49a1 is one of three K57 RATIFIED Bridge Objects in BST.

7.4 1/rank universality (T1430)

Task completed: $L(E, 1)/\Omega = 1/\text{rank}$ for 49a1, where L is the L-function and Ω the real period.

T1430 universality: extension from 49a1 to all 7 Millennium problems + Four-Color theorem. The 1/rank pattern characterizes broad classes of mathematical structures.

This is BST's "Crown Jewel" in the elliptic-curve / arithmetic-geometry sector.

7.5 Cremona 27a1 and Cremona 121a1

Cremona 27a1: $Y^2+Y = X^3$. CM by $\mathbb{Q}(\sqrt{-3}) = \mathbb{Q}(\sqrt{-N_c})$. Anchored as K-audit candidate (Task #186 completed).

Cremona 121a1: $Y^2+Y = X^3-X^2-7X+10$. CM by $\mathbb{Q}(\sqrt{-11})$. 4th Bridge Object candidate (Task #245 completed at 3.5/4 audit).

Together with 49a1: the trio at BST primary discriminants $\{-N_c, -g, -c_2\text{-related}\}$.

7.6 K-audit anchors

- Heegner-Stark 1952-67 (L1 ESTABLISHED)
- K75 audit: BST anchors on Stark small-primary subset
- K57 RATIFIED: 49a1 as Bridge Object
- T1430: 1/rank universality
- Tasks #186, #245: Cremona 27a1, 121a1

Level 3 — 5th-grader accessibility

Heegner numbers: imaginary quadratic fields $\mathbb{Q}(\sqrt{-d})$ with class number 1. Heegner-Stark-Baker proved there are exactly 9: $d \in \{1, 2, 3, 7, 11, 19, 43, 67, 163\}$. BST anchors on the small-primary subset $\{-3, -7, -11\}$ — these match BST primary structure. Cremona 49a1 is BST's canonical elliptic curve: every invariant is BST primary (conductor $g^2 = 49$, $j = -(3 \cdot 5)^3$, CM by $\mathbb{Q}(\sqrt{-7})$). 1/rank universality (T1430): L-function value / period ratio = 1/rank for 49a1, extended to all 7 Millennium problems + Four-Color theorem.

What comes next

Chapter 8 develops Monster, Moonshine, and Supersingular primes.

Where to look this up

- Heegner 1952, Stark 1967, Baker 1966
- Cremona, *Algorithms for Modular Elliptic Curves*
- BST: K57, K75, T1430